

Basic Principles of Medicine 2

Module: Neuroscience and Behaviour

Clinical Neuroimaging:

Directed Learning Activity

To be studied after Lecture

Organization of the Nervous

System

Images from *Neuroanatomy: An Atlas of Structures, Sections and Systems 10th Ed.* D.E. Haines © 2019 Lippincott Williams and Wilkins unless otherwise stated

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Recommended Reading

1. Siegel and Sapru, Essential Neuroscience, Fourth Edition, 2019, Chapter 27, Vascular Syndromes.

Part III:

Magnetic Resonance Imaging (MRI)

To be studied after Lecture
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System

Learning Objectives

SOM.CS.I.BPM2.1.NSCI.0048 Describe the physics, diagnostic use and contraindications/limitations of X-ray-based techniques (excluding CT).

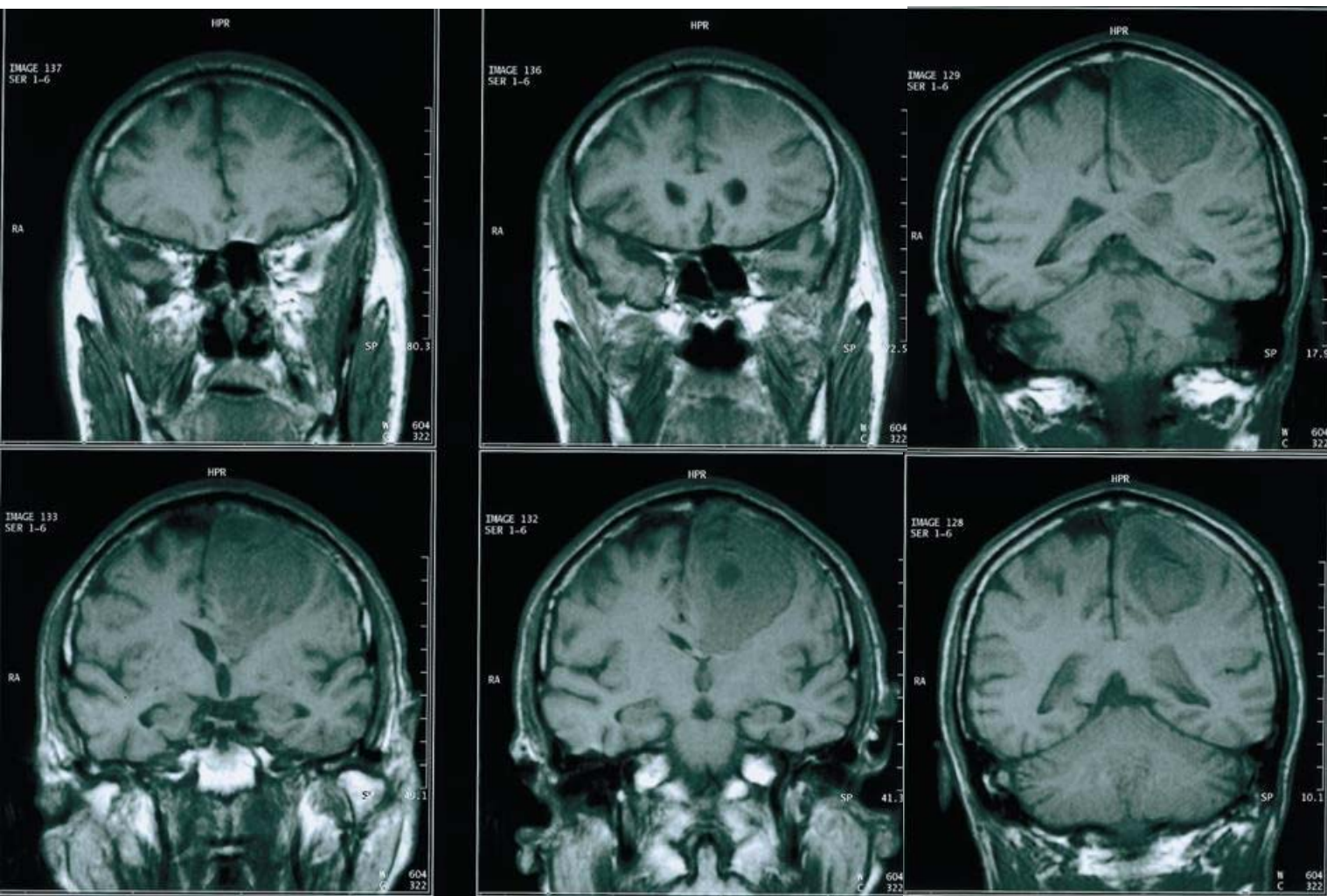
SOM.CS.I.BPM2.1.NSCI.0049 Describe the physics, diagnostic use and contraindications/limitations of CT.

SOM.CS.I.BPM2.1.NSCI.0050 Describe the physics, diagnostic use and contraindications/limitations of MRI.

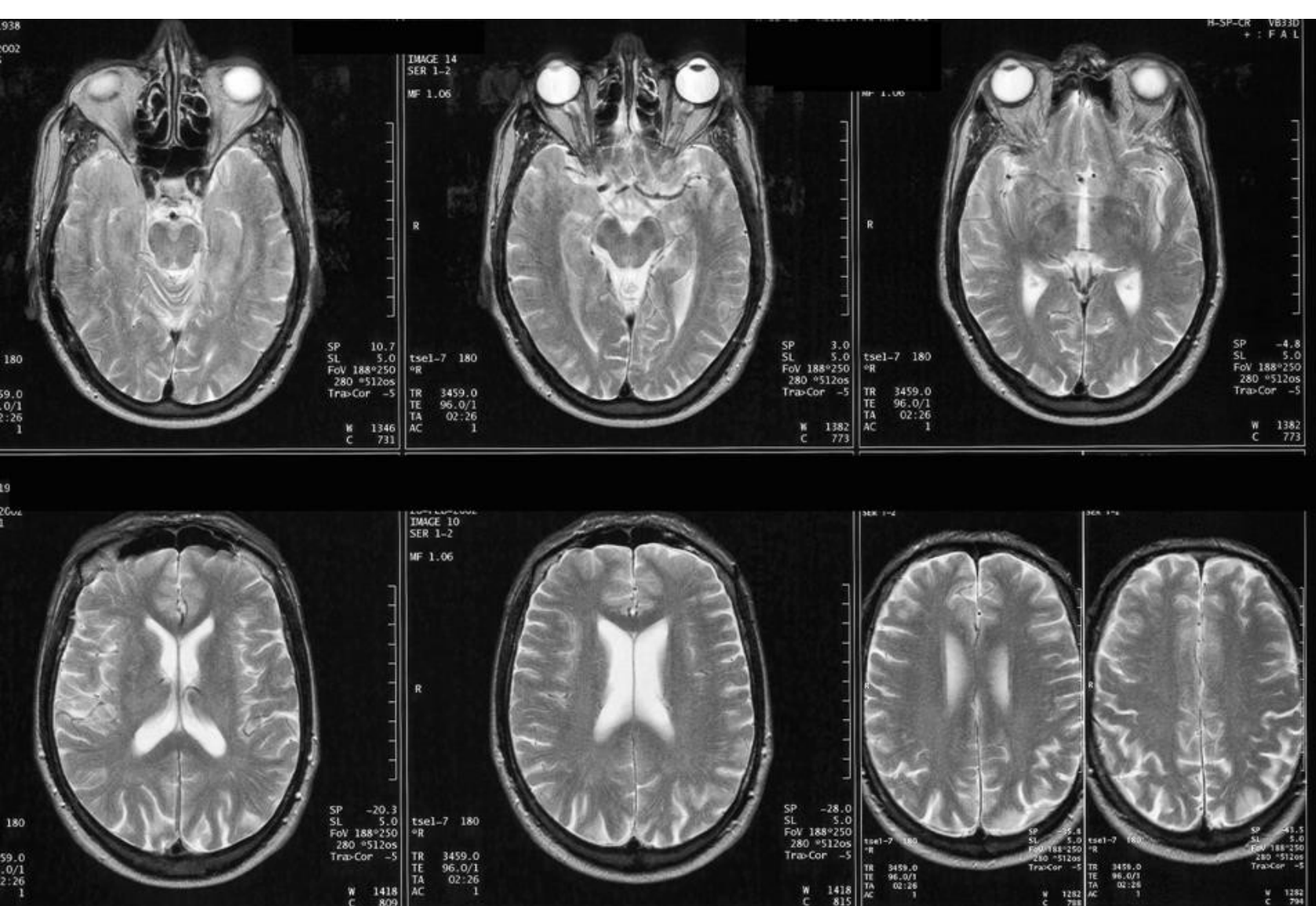
Basic Magnetic Resonance Imagery

- Magnetic moments of hydrogen protons aligned in strong magnetic field
- Radio waves change alignment and synchrony of spin, with protons subsequently relaxing to their prior state
- Relaxation yields signals that generate an image
- T1 signals (strong in fatty tissue) and T2 signals (strong in watery tissue) revealing distinct aspects of anatomy and pathology

T1 MRI



- T1 MRI differentiates gray-white and shows structure in detail (CSF is dark)
- Infarctions are visible

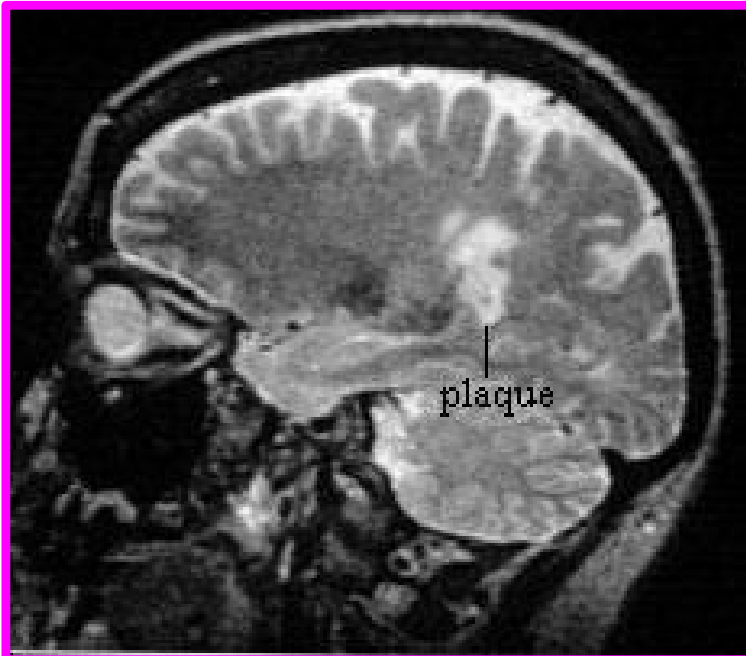


T2 MRI

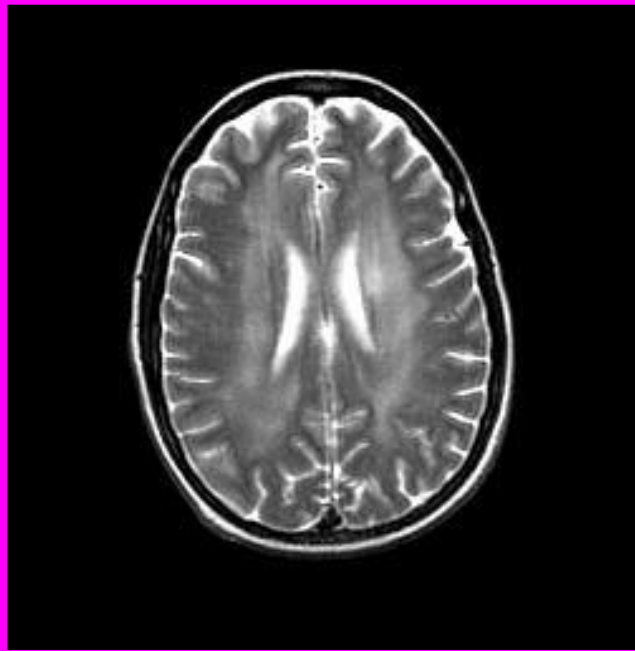
- CSF is white
- Weak differentiation of white and gray
- Shows edema and hence neuropathology

T2-weighted MRI

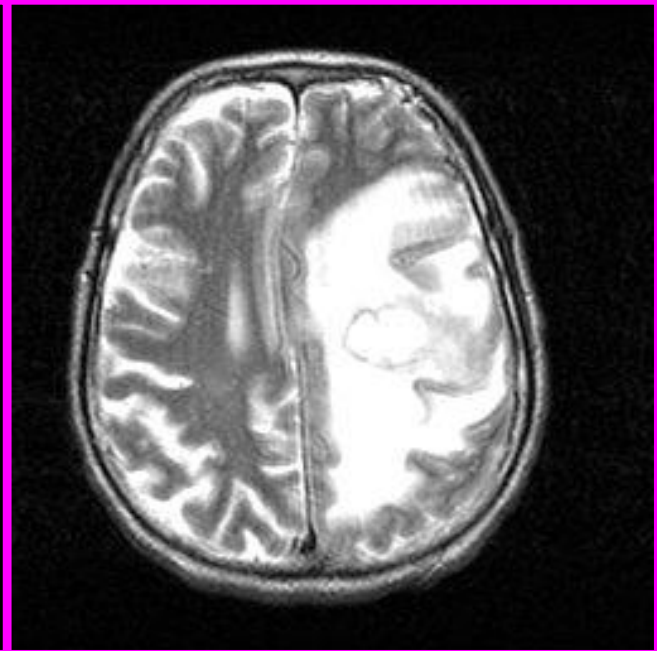
- Sensitive to pathology.



Multiple Sclerosis*



AIDS



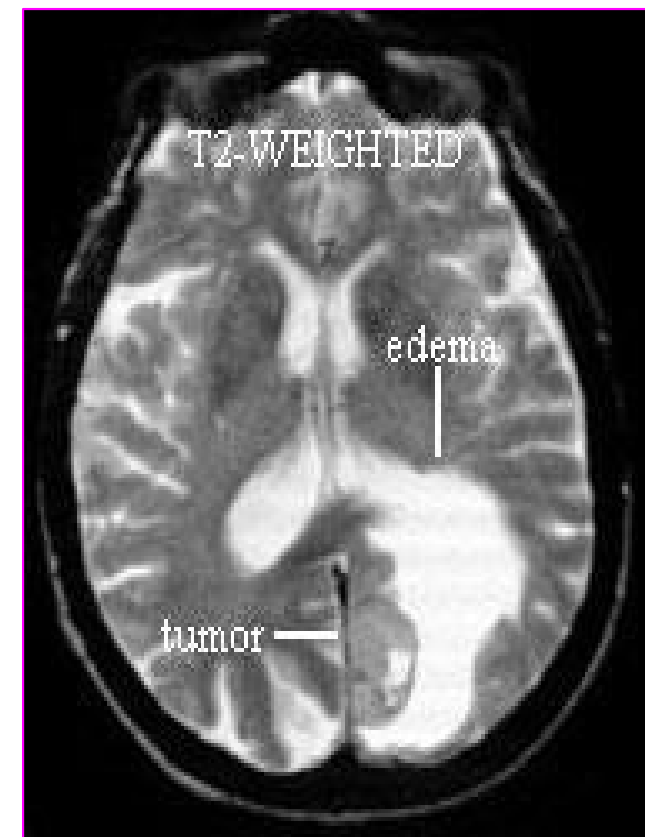
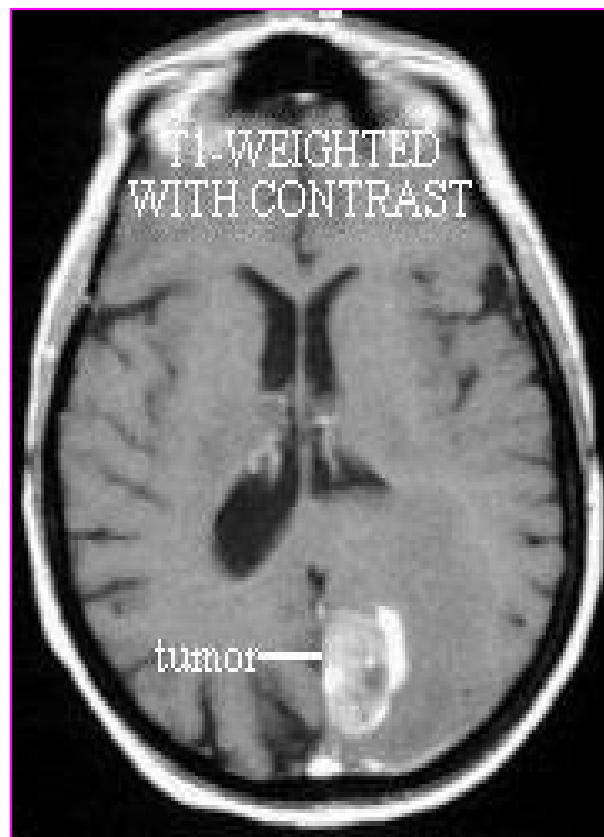
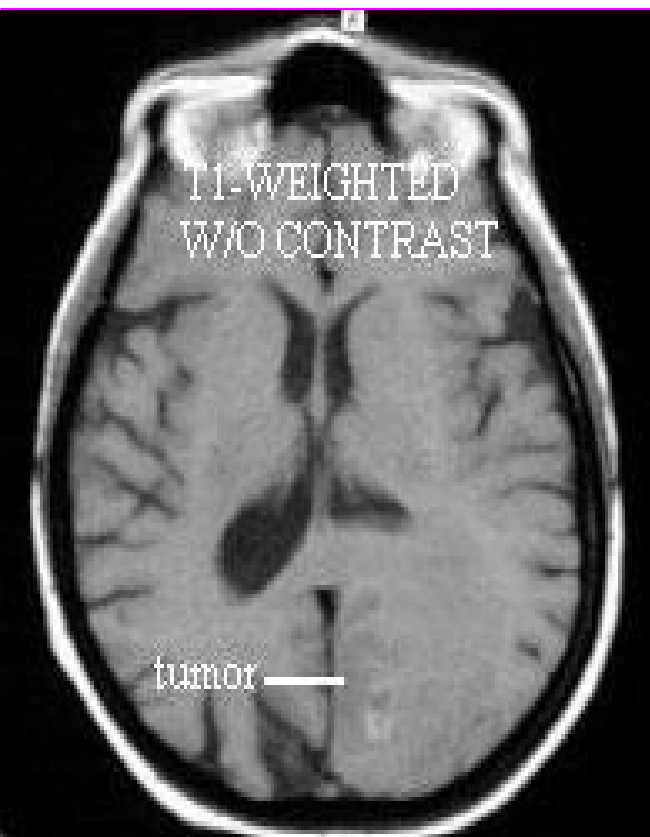
Abscess

*A specialized MRI variant called FLAIR, which darkens the CSF, may better reveal periventricular lesions from multiple sclerosis than a standard T2-weighted scan

Comparison of T1- and T2-weighted Scans in Tumor Detection

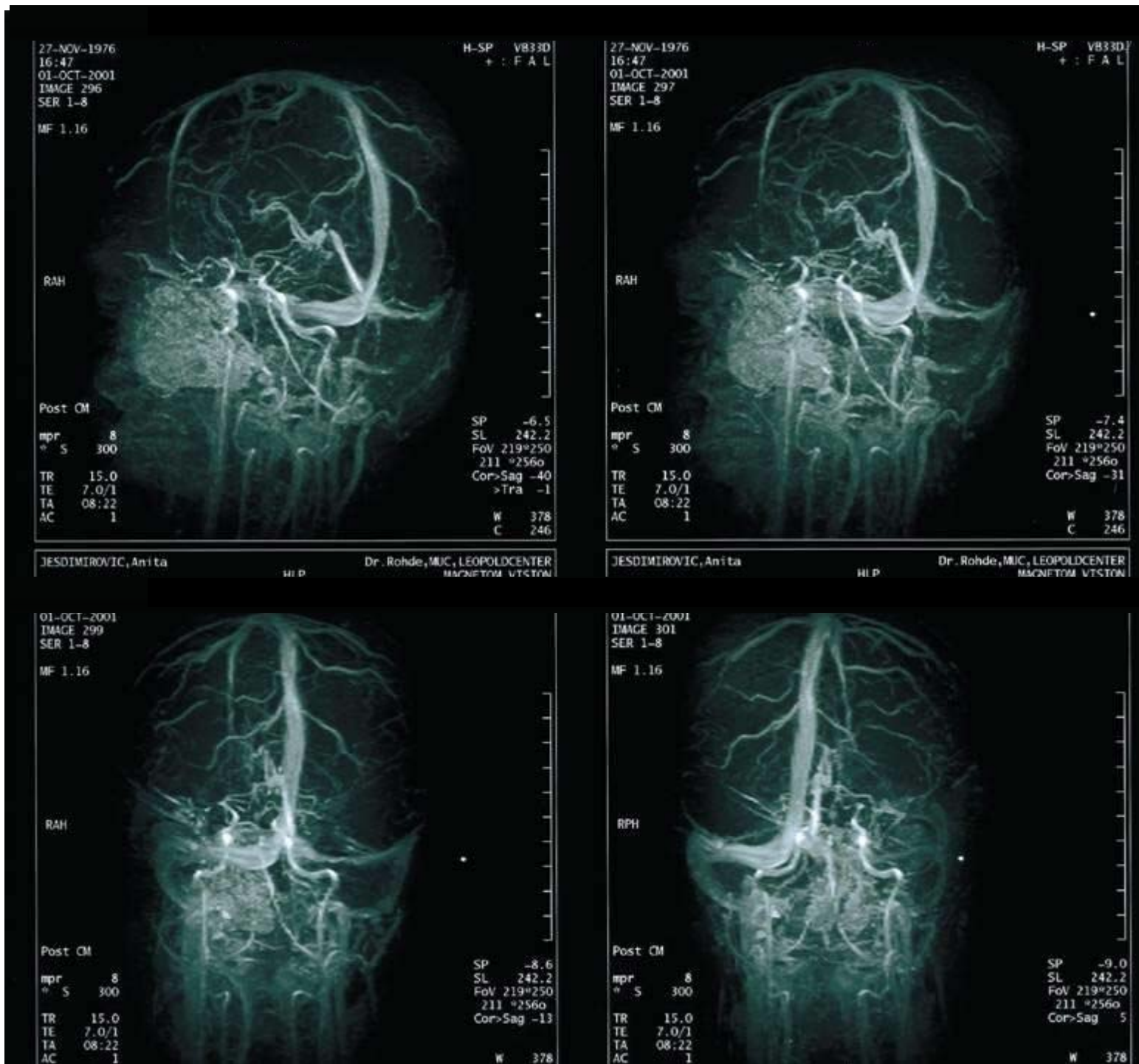
- T1-weighted MRI, without contrast, is often poor at localizing boundaries of tumors in brain.
- A contrast agent dramatically improves the capacity of the T1-weighted MRI to reveal the boundaries of the tumor, which are associated with compromise of the blood-brain barrier.
- T2-weighted MRI poorly demonstrates tumors but can reveal inflammatory reactions in surrounding tissue.

Comparison of T1- and T2-weighted Scans in Tumor Detection



MR Angiography

- Contrast agent (gadolinium) is introduced to the vascular system to reveal normal and abnormal blood vessels (e.g., expressing stenoses or aneurysms)



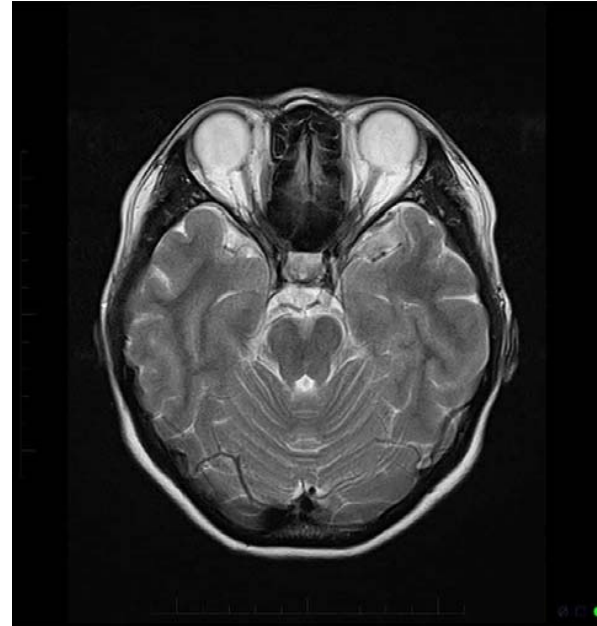
MR Angiography

MRI: Advantages and Disadvantages

- Advantages
 - Excellent structural resolution
 - Sensitive to edema
 - No ionizing radiation
 - Relatively common and accessible
- Disadvantages
 - More expensive than CT
 - Cannot be used when paramagnetic metals in body (watch also for cardiac pace-makers)

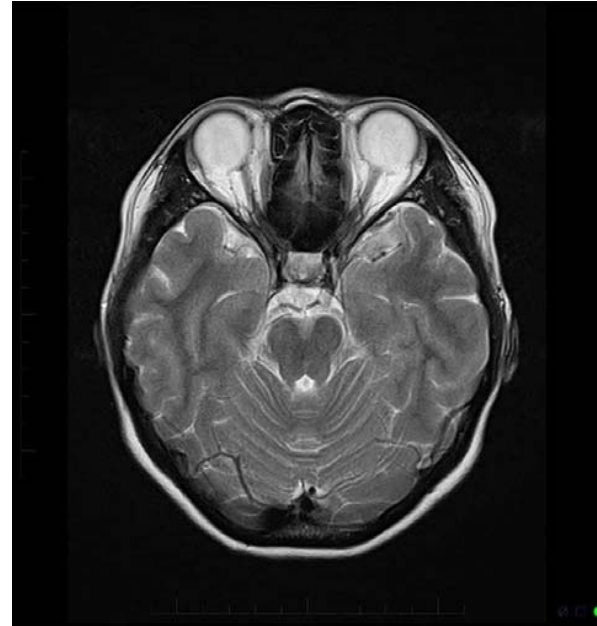
Which of the following neuroimaging techniques most likely yielded the image below?

- A. X-ray
- B. CT
- C. DSA
- D. MRA
- E. MRI (T1)
- F. MRI (T2)



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- D. MRA
- E. MRI (T1)
- ☺ F. MRI (T2)



The well-defined structure indicates MRI. Note that the fluid of the intraventricular, subarachnoid and intraocular spaces is white, indicating a T2 signal.